

University of Dhaka

Department of Computer Science and Engineering
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CSE 4113: Internet Programming Lab Project

DUCSU eVote

Functional and Non-Functional Requirements

A Secure and Privacy-Preserving Online Voting Platform for DUCSU Elections

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1. System Overview

DUCSU eVote is a web application for running Dhaka University Central Students' Union (DUCSU) style elections online. The Election Commission sets up the election and its 28 central posts, candidates register individually or as part of a panel, and eligible students log in with their registration number and a one-time password to cast a secret ballot from a phone or laptop. When a vote is submitted, the system captures a photo of the voter through the device camera as proof that the student personally cast the vote. The system shows live turnout while voting is open and calculates and publishes results automatically once voting closes. Its core principle is that the system records *that* a student has voted, including photo evidence of who cast the vote, but stores no data that shows *whom* they voted for.

2. Functional Requirements

ID	Requirement	System Functionality
FR-01	Voter Authentication	Voters log in with their registration number and a one-time password (OTP) sent to their registered university email. Each OTP expires after 5 minutes, can be used only once, and repeated failed attempts trigger a temporary cooldown.
FR-02	Role-Based Access Control	The system supports separate Voter, Candidate, Election Commission and Observer roles. Election Commission and Observer users log in with email and password, every page and action is restricted by role, and sessions end on logout or after a period of inactivity.
FR-03	Election Setup and Scheduling	The Election Commission creates an election with a title, start time and end time, and defines each position with its display order and maximum number of selections (for example, 1 for Vice-President and 13 for member posts). Voting opens and closes automatically at the scheduled times, and the election status is shown to all users.
FR-04	Voter List Management	The Election Commission imports the voter list from a CSV file and marks voters as eligible or ineligible. Positions, candidates and the voter list cannot be changed once voting has started.
FR-05	Candidate Nomination and Panels	Students submit a nomination for one position and upload a photo, short biography and manifesto. Candidates can create or join a panel or stand as independents, and panels can publish a name, slogan, logo and joint manifesto.
FR-06	Nomination Review	The Election Commission approves or rejects nominations, giving a reason for each rejection. A candidate's profile becomes public only after approval.

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FR-07	Candidate Information and Panel Comparison	Voters can browse approved candidates grouped by position, view each candidate's profile and manifesto, search and filter candidates, and compare two panels side by side, issue by issue.
FR-08	Step-by-Step Ballot	The ballot presents one position at a time with a progress indicator. It prevents selecting more candidates than a position allows, both in the browser and on the server, allows a position to be left blank, and lets voters open a candidate's profile without losing their selections.
FR-09	Ballot Review and Submission	Before submitting, voters see a review screen listing all their selections, can go back to change them, and must confirm the final submission. On confirmation, the voter's photo is captured through the device camera (FR-19) before the ballot is submitted.
FR-10	One Person, One Vote	Only eligible voters who have not yet voted can vote, and only while voting is open. Recording the ballot, saving the voter's photo, marking the voter as voted and creating the receipt happen in one atomic database transaction with the voter's record locked, so simultaneous requests cannot produce a second vote. Further attempts are rejected with a clear message.
FR-11	Secret Ballot Storage	Ballots are stored without any voter ID, registration number or timestamp, are identified by random UUIDs rather than sequential numbers, and are never written to application or server logs. The system records only whether a voter has voted, not when. As an extension, candidate choices can be encrypted with the Election Commission's public key and decrypted only at result calculation.
FR-12	Vote Receipt	After voting, each voter receives a random receipt code that they can copy or download. Receipt codes are stored with no link to voters or ballots. After voting closes, the full list of receipt codes is published with a search box so voters can confirm their ballot was counted. Receipts never show the voter's choices.
FR-13	Live Turnout Monitoring	While voting is open, the system shows the number of registered voters, votes cast, turnout percentage and turnout by hall. No candidate's vote count is calculated or displayed until voting closes.

ID	Requirement	System Functionality
FR-14	Result Calculation and Publication	The Election Commission can calculate results only after the voting end time. The system counts votes for every position, identifies the winners up to each position's number of seats, flags ties for the Election Commission to resolve, and publishes results with a bar chart for each position. It also shows that the numbers of ballots, receipts and voters marked as voted are equal.
FR-15	Observer Access	Observers have read-only access to live turnout, system status, the audit log and its integrity check, without access to any personal voter data.
FR-16	Tamper-Evident Audit Log	Every administrative action, such as approving a candidate, opening voting or calculating results, is recorded with the actor, action, time and IP address. Each entry stores a hash of the previous entry, an integrity check reports any modified or deleted entry, and log entries cannot be edited or deleted through the application.
FR-17	Email Notifications	The system sends OTP codes by email, notifies eligible voters when voting opens, reminds voters who have not yet voted before voting closes, and notifies users when results are published.
FR-18	Language Support and Transparency	Voter-facing pages offer a Bangla/English language switch. The system provides a plain-language page explaining how each vote is kept secret, explains on the voting page why candidate counts are hidden until voting closes, and links to the project's public source code repository.
FR-19	Voter Photo Verification	When a voter confirms their ballot, the system asks for camera permission, shows a live preview and captures a photo of the voter's face. The ballot cannot be submitted without a photo. The photo is stored with the voter's record only, so it proves that the student personally cast a vote but is never linked to their ballot or choices. Photos can be viewed only by the Election Commission to resolve a dispute, such as a student denying that they voted, and every viewing is recorded in the audit log. Photos are deleted automatically once the election's dispute period ends.

3. Non-Functional Requirements

ID	Quality Requirement	Expected Standard
NFR-01	Usability	The interface shall be mobile-first and fully usable on screens 360 pixels wide, and a voter shall be able to complete the full 28-post ballot in under 5 minutes.
NFR-02	Performance	Pages shall load within 3 seconds on a typical 4G mobile connection, a ballot submission shall complete within 2 seconds under normal load, and live turnout figures shall refresh at least every 10 seconds.
NFR-03	Scalability	The system shall handle at least 500 simultaneous voters, verified by a load test, without errors or lost votes.
NFR-04	Reliability and Availability	The system shall be available throughout the voting window with 99% uptime. If any part of a ballot submission fails, the whole transaction shall roll back and the voter shall be asked to try again, so that no accepted ballot is ever lost. The database shall be backed up before voting opens and after it closes.
NFR-05	Security	All traffic shall use HTTPS. Admin passwords shall be hashed with bcrypt and OTPs stored only as hashes, session tokens shall be kept in HTTP-only secure cookies, and OTP requests and login attempts shall be rate-limited. All input shall be validated on the server, database queries shall be parameterised, and secret keys shall be kept in environment variables rather than source code.
NFR-06	Ballot Secrecy and Privacy	It shall be impossible to link a voter to their ballot from any stored data, including database tables, backups, logs and voter photos. Used or expired OTPs shall be deleted, observers shall have no access to personal voter data, and the demonstration shall use dummy data only. Voter photos shall be stored with random file names, with image metadata such as capture time and location removed, shall be accessible only to the Election Commission, and shall be deleted after the dispute period. Voters shall be informed before voting that a photo will be taken and why.
NFR-07	Integrity and Transparency	The numbers of ballots, receipts and voters marked as voted shall always be equal, the audit log shall reveal any tampering, and the source code shall be published in a public repository.
NFR-08	Accessibility and Compatibility	Text and controls shall meet WCAG 2.1 AA colour contrast, and the ballot shall be operable by keyboard. The system shall work on current versions of Chrome, Firefox, Safari and Edge on Android, iOS, Windows and macOS, including camera access for voter photo capture.

ID	Quality Requirement	Expected Standard
NFR-09	Maintainability	The backend shall expose a documented REST API separate from the frontend, settings such as OTP expiry and rate limits shall be configurable without code changes, and the voting transaction and secrecy rules shall be covered by automated tests, including simultaneous double-vote attempts.
NFR-10	Deployability	The system shall be deployed on a public host and be fully usable online.

4. Feature Flow Summary

Election Setup	The Election Commission creates the election, defines the 28 positions and imports the voter list
Nomination	Candidates register with their profile and manifesto, form or join panels, and are approved by the Election Commission
Voter Login	Eligible students log in with their registration number and an email OTP
Voting	Voters read candidate information, complete the step-by-step ballot and review it; a photo of the voter is captured on submission, and the voter receives a receipt code
Monitoring	Live turnout is shown while voting is open; observers watch system status and the audit log
Results and Verification	Results are calculated after voting closes and published with charts, and voters check their receipt codes against the published list